

Deepak Mallampati

APPLIED AI / LLM ENGINEER - HEALTHCARE & REGULATED SAAS

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Kent, OH - US-based, F-1 OPT

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building production LLM and RAG systems for regulated banking and healthcare environments. Hands-on with OpenAI, Anthropic, and AWS Bedrock APIs; vector databases (FAISS, Pinecone, Weaviate); embeddings and semantic search. Built evaluation, observability, and hallucination-guardrail frameworks (RAGAS, BERTScore) that catch regressions pre-deployment. Production track record: **40% hallucination reduction**, **92% retrieval precision**, p95 latency 1.5s → 800ms, and **99.9% uptime** at enterprise scale across HIPAA, SOC 2, and PCI DSS environments.

SKILLS

LLM API Integration (OpenAI, Anthropic, AWS Bedrock) • RAG Architecture & Vector Search • Embeddings & Semantic Caching • Model Evaluation & Observability (RAGAS, BERTScore) • Hallucination Guardrails • AWS Cloud-Native Architecture • HIPAA / PHI Handling & Audit Logging • Latency & Inference-Cost Optimization • Python Backend + FastAPI • Production LLM Deployment

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety | Hugging Face

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Cross-Encoder Re-ranking | Semantic & Hybrid Search | Document Chunking | Semantic Caching

MLOps & Cloud: AWS (Bedrock, SageMaker, EC2, S3, Lambda), Azure (OpenAI, AI Services), GCP Vertex AI | Docker, Kubernetes | Terraform | MLflow, W&B | Observability

Evaluation & Monitoring: RAGAS, BERTScore, Custom domain metrics | A/B Testing | Drift Detection | Latency/Accuracy Dashboards

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

Languages & Tools: Python, FastAPI, JavaScript/TypeScript, Java, SQL | PostgreSQL, MongoDB, Redis | Spark, Kafka | Git, Jira

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer - USA

- Built production **LLM API integrations** across OpenAI, Anthropic, and Bedrock - agentic conductor-subagent orchestration with context-scoped subagents, cutting token consumption by **42%** while sustaining multi-step task accuracy.
- Engineered **RAG pipelines** over large-scale healthcare document corpora with dense retrieval + cross-encoder re-ranking, reducing hallucination by **40%** and lifting precision to **92%** on regulated documents.
- Stood up the **LLM evaluation and observability** framework (RAGAS, BERTScore, custom domain metrics) with latency + accuracy dashboards - catching regressions pre-deployment.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets with K-anonymity, L-diversity, and Laplace differential privacy - enabling compliant analytics with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled orchestration and exponential-backoff retries, cutting failed production runs by **60%**.
- Optimized **inference cost and latency** via prompt compression, semantic caching, and model routing - sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern - USA

- Built and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability across **25+ AI agents**.
- Evaluated **GPT-4o, Claude Sonnet, and Gemini** on internal benchmarks - improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered **safeguards and content controls** for policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Built **12 APIs and microservices** powering new AI capabilities - p95 latency 1.5s → **800ms**; accelerating feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & GenAI Research Assistant - USA

- Built a **privacy-preserving ML pipeline** on hospital-readmission data - k-anonymity (k=3), l-diversity (l=2), Laplace DP across four ϵ levels, on 8 clinical features.
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline AUC** (0.689 vs 0.696); weighted-F1 0.895.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on NVIDIA H100 with a 361-example instruction dataset; authored IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee - India

- Developed supervised **regression and classification** models on industrial operational data - equipment-failure prediction, **anomaly detection**, capacity forecasting.
- Engineered features and end-to-end data preprocessing on raw sensor and operational pipeline data, producing model-ready training pipelines.
- Fine-tuned **BERT/RoBERTa** for entity extraction + text classification on financial datasets - **89% F1** on custom NER.

Suvidha Foundation

Jan 2022 - Mar 2022

ML Intern - India

- Designed an end-to-end **video summarization + highlight-generation** pipeline (speech-to-text + transformer abstractive summarization + timestamp-aligned clip extraction).
- Deployed across **5,000+ videos**, automating highlight selection and reducing manual editing effort by **70%**.

PROJECTS

MangaLens

Shipped Chrome extension delivering real-time AI translation of manga/webtoons across 29+ sites. Production product on the Chrome Web Store; OCR + LLM translation in a single inline action.

career-ops

Autonomous AI-driven job-search pipeline: scans listings, scores fit, generates tailored applications overnight without supervision. Production automation across LLM evaluation, RAG, PDF generation.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy on clinical readmission data; reduced re-identification risk from 3.38% to 0.32% while retaining 99% of baseline AUC.

Story Director LLM

Qwen3-27B fine-tuned via 4-bit QLoRA on H100 for controllable long-form narrative generation, paired with an automated video rendering pipeline.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70 / 4.0